MAP COLORING USING CSP

Aim

To write a Python program to implement the Map Coloring problem using Constraint Satisfaction Problem (CSP).

Objective

To color a map such that no two adjacent regions have the same color.

Algorithm

1. Start.
2. Define the map and neighboring regions.
3. Assign colors to each region.
4. Check whether adjacent regions have different colors.
5. If the coloring is valid, display the result.
6. Stop.

Source Code

```
colors = {

    "A": "Red",

    "B": "Green",

    "C": "Blue",

    "D": "Red"

}


graph = {

    "A": ["B", "C"],

    "B": ["A", "C", "D"],

    "C": ["A", "B", "D"],

    "D": ["B", "C"]

}


valid = True


for node in graph:
```

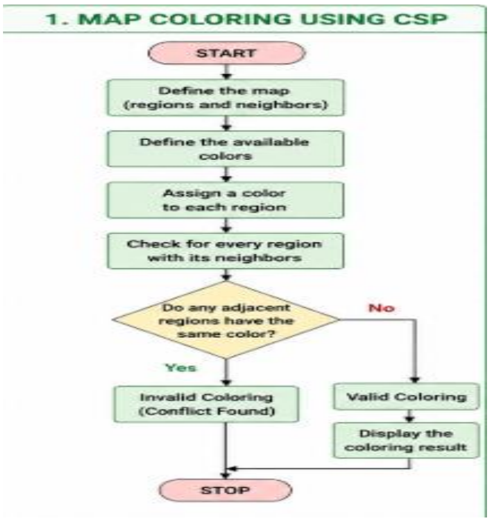
```
    for neighbour in graph[node]:
        if colors[node] == colors[neighbour]:
            valid = False

if valid:
    print("Valid Coloring")
    print(colors)
else:
    print("Invalid Coloring")
```

Sample Output

```
Valid Coloring
{'A': 'Red',
 'B': 'Green',
 'C': 'Blue',
 'D': 'Red'}
```

FLOWCHART:



Result

Thus, the Python program for Map Coloring using CSP was executed successfully.